Lab 2: Knowledge Is Power

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1. Introduction

This portion of the Software Requirements Specification (SRS) introduces the Knowledge Is Power (KIP) project and describes its general design. It follows the IEEE 830-1998 structure used by Team Crystal for Lab 2. Detailed requirements for the individual system features appear in the team's Section 3.

1.1 Purpose

The purpose of this SRS is to document what the KIP system is expected to do, who will use it, and the conditions that affect its development. It gives the development team, course instructor, and other reviewers a common reference for designing, implementing, and testing the software. The document describes the product at a high level before the team lists more specific, testable requirements for each system feature.

1.2 Scope

Knowledge Is Power (KIP) is a web-based legal literacy platform designed primarily for young people ages 16 to 19. Young adults face decisions involving driving, online conduct, work, and other legal situations even when they have not received much practical instruction about those laws. KIP aims to help them recognize when a decision has legal consequences and understand why those consequences matter (Team Crystal, 2026).

Instead of relying on definitions alone, KIP will use short quizzes, interactive scenarios, immediate feedback, and references to official legal sources. Its initial prototype focuses on Virginia traffic law and copyright issues involving social media. The larger product could eventually support additional legal topics, but KIP is meant to supplement education rather than replace formal legal instruction or provide legal advice.

1.3 Definitions, Acronyms, and Abbreviations

The following terms are taken from the glossary used in Team Crystal's Lab 1:

CDC: Centers for Disease Control and Prevention

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

UX: User Experience

1.4 References

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1.5 Overview

Section 1 explains the purpose and scope of this SRS and provides definitions and references. Section 2 describes KIP's product perspective, main functions, users, constraints, and assumptions. Section 3 of the team's SRS provides the detailed system requirements, including the purpose, stimulus and response, and functional requirements for each of KIP's 15 system features.

2. Overall Description

This section summarizes how KIP is organized and how users are expected to interact with it. The detailed behavior of the features is defined separately in Section 3 of the team SRS.

2.1 Product Perspective

KIP is a browser-based learning application that combines an interactive interface, application logic, and saved user data. The site includes navigation, a dashboard, a game browser, account access, settings, and an administrator panel. Once a game begins, the game window displays legal scenarios and accepts the learner's decisions, while the game engine manages the order of scenarios and stages.

The remaining parts of the system support the learning process. Feedback evaluates responses, Progress Tracking records results, and Curation uses performance information to suggest future practice. The Legal Reference Algorithm is intended to identify changes in legal sources. The team's Section 3 also includes data initialization and user-profile and match-metric features that support the dashboard and the storage of progress information. KIP is designed for access through a modern web browser, without special software installed by the learner.

The current project uses web technologies with a Java-based application layer and JHipster-supported database setup. The team's detailed requirements describe Liquibase and custom seed data for prototype testing. These implementation details support the planned features but do not mean that every feature is already complete.

2.2 Product Functions

KIP's major functions are organized around the following activities (Team Crystal, 2026):

- **Accounts and navigation.** The Account Portal supports access to registered learner accounts, while the Navbar, Browser, and Dashboard help users find games and see recommendations and recent activity. User Settings and the sitewide theme provide a consistent interface.

- **Interactive games.** The “Fact or Fiction?” game checks common legal beliefs using short responses. “Choose the Outcome” presents longer situations with different decisions and consequences. The Game Window displays each scenario, and the Game Engine manages responses and advancement through the game.
- **Feedback and legal references.** The Feedback system explains the results of a learner’s responses and passes relevant data to Progress Tracking. Legal-source features help connect learning material to accurate references and identify material that may need review when laws change.
- **Progress tracking and recommendations.** KIP records performance for concepts, topics, and games. The Curation system uses those results to recommend areas where a learner is improving or needs additional practice, while the Dashboard presents relevant activity information.
- **Administration and supporting data.** Administrative functionality helps authorized users manage the platform. Data initialization and profile and match metrics provide the information needed for testing, account summaries, and game history.

For the prototype, the learning content is limited to selected Virginia traffic and copyright examples. Features described here represent the intended system behavior; the team’s Section 3 defines the more specific requirements that will be evaluated.

2.3 User Characteristics

KIP is designed for learners who have basic experience using websites but may have little or no background in legal terminology. Its main audience is young adults ages 16 to 19. The intended user types are:

- **Registered learners.** Users who sign in to play games, review saved activity, track their progress, and receive recommendations based on their performance.
- **Guest learners.** Visitors who may be able to try a limited part of KIP without creating an account. Guest access is part of the planned experience, rather than a promise of saved progress.

- **System administrators.** Authorized users who maintain platform settings and review or update learning content, including material flagged as potentially outdated.

Parents, teachers, and school administrators may use KIP as a supplemental educational resource, but the project does not define separate application permissions for those groups.

2.4 Constraints

KIP is intended to run in a modern web browser on desktop and mobile devices and requires internet access. The prototype is limited to selected Virginia traffic and copyright-law content, so it will not provide comprehensive legal coverage. KIP must also avoid presenting its educational feedback as professional legal advice. The team's technology choices and the need to keep legal references accurate affect implementation and maintenance.

2.5 Assumptions and Dependencies

KIP assumes users have a compatible browser and reliable internet access. The accuracy of the lessons depends on official legal sources and on the team reviewing material when those sources change. Personalized recommendations also depend on the game engine, feedback, account data, and progress tracking working together. These dependencies must be connected and tested for the prototype to demonstrate the intended learning cycle.